Example 3

Analysis of CO_2 Emissions

Goals

1. Compute CO_2 per 1000 people per year + region + source
    1. Goal: Create a longitudinal line graph
2. Compute Total CO_2 per 1000 people per year + region
    1. Goal 1: Compute the top 10 overall combinations (year + region)
    2. Goal 2: Compute the top 3 regions per year for 2018+

Inspecting the raw data

Country Name	Country Code	Series Name	Series Code	2025 (YR202...	2024 (YR20...	2023 (YR20...	2022 (YR20...
Africa Eastern and Southern	AFE	Population, total	SP.POP.TOTL	788844284	769280888	[illegible]	731821393
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.MT.CE.AR5	..	634.4063	619.3657	618.5414
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PC.CE.AR5	..	0.837679788408201	[illegible]	0.858131540848484
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.AG.MT.CE.AR5	..	2.8442	[illegible]	2.6826
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.BU.MT.CE.AR5	..	41.7037	[illegible]	41.177
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.FE.MT.CE.AR5	..	77.7873	[illegible]	74.5494
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IC.MT.CE.AR5	..	78.5775	[illegible]	73.9143
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IP.MT.CE.AR5	..	42.5011	[illegible]	39.1414
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PI.MT.CE.AR5	..	264.9669	250.9278	260.1308
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.TR.MT.CE.AR5	..	126.023	127.844	126.9431
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.WA.MT.CE.AR5	..	0.0034	0.0033	0.0032
Africa Western and Central	AFW	Population, total	SP.POP.TOTL	532809933	520655398	508318102	496366058
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.MT.CE.AR5	..	261.934	251.6472	259.3996
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PC.CE.AR5	..	0.503085151918467	0.495058505707121	0.52259737711558
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.AG.MT.CE.AR5	..	1.514	1.4556	1.4155
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.BU.MT.CE.AR5	..	12.426	11.9931	12.3841
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.FE.MT.CE.AR5	..	34.9597	32.4378	32.9127
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IC.MT.CE.AR5	..	21.7507	21.6229	22.5579
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IP.MT.CE.AR5	..	33.7165	33.2706	34.4196
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PI.MT.CE.AR5	..	45.5711	42.9397	44.1713
Africa Western and Central	AFW	[illegible]	EN.GHG.CO2.TR.MT.CE.AR5	..	111.9837	107.9158	111.5267
Africa Western and Central	AFW	[illegible]	EN.GHG.CO2.WA.MT.CE.AR5	..	0.0124	0.0119	0.0115
Arab World	ARB	Population, total	[illegible]	503356133	492612632	482105978	471352066
Arab World	ARB	[illegible]	[illegible]	..	2229.854	2171.3007	2106.1097
Arab World	ARB	[illegible]	[illegible]	..	4.57571631886073	4.55256379383056	4.51655911861293
Arab World	ARB	[illegible]	[illegible]	..	3.9359	3.9489	4.017

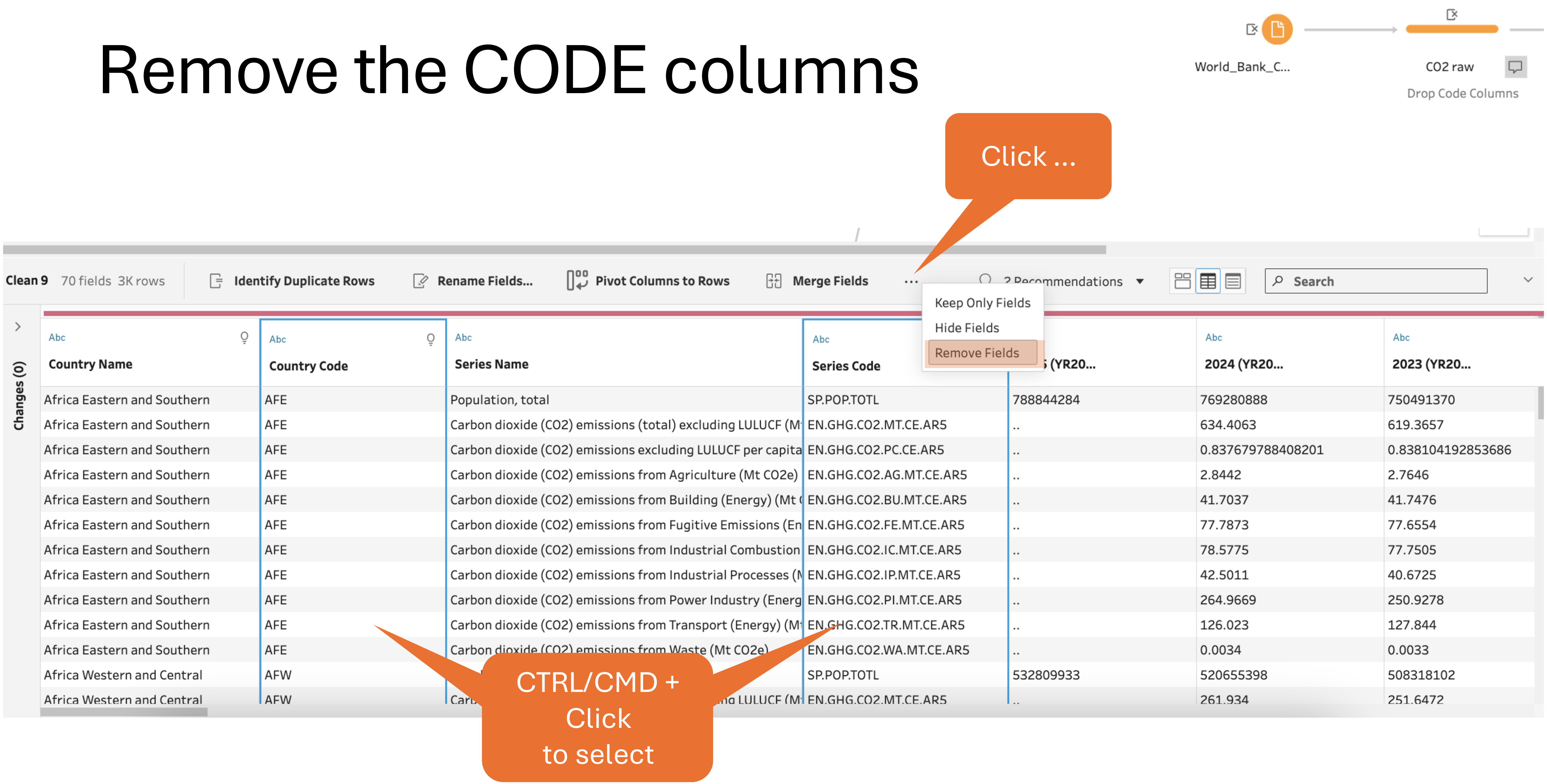
Drop

Drop

Data in NAME
→ Stack

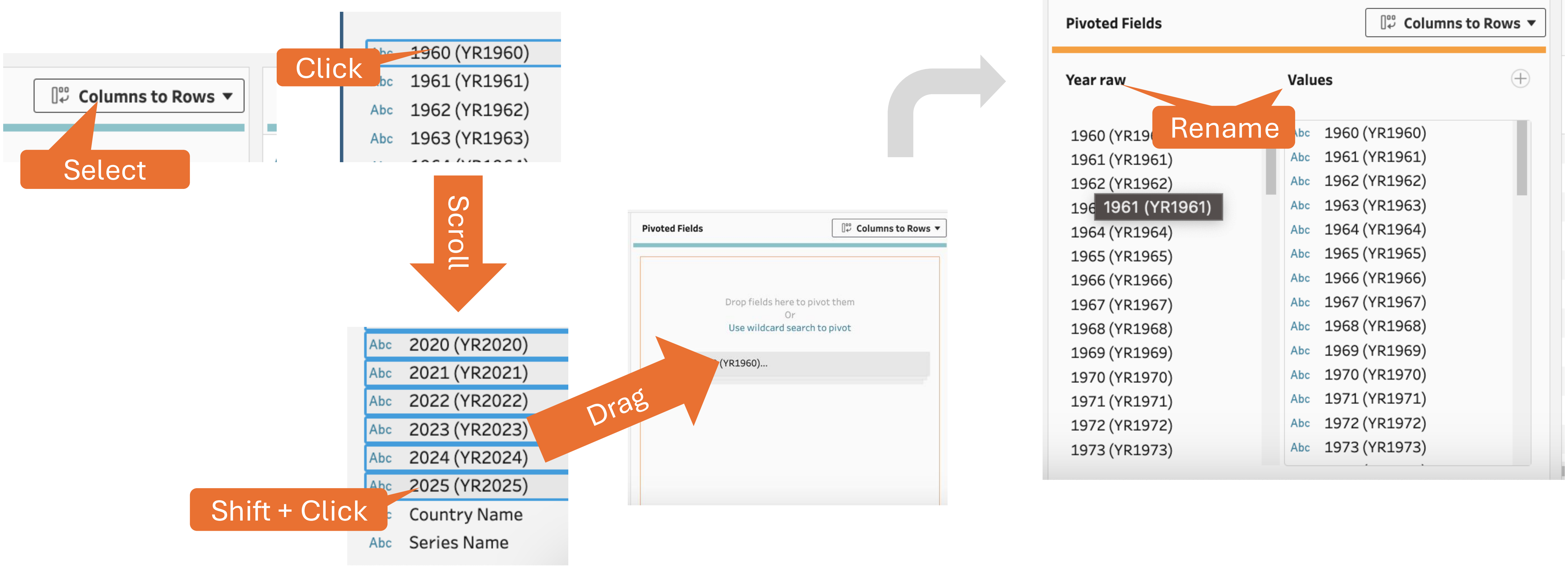
Variable in COLUMN
→ Unstack

Remove the CODE columns



Stack the YEAR columns

Pivot → Columns to Rows



Goal: Clean up the VARIABLE names

Solution: Click and edit the labels

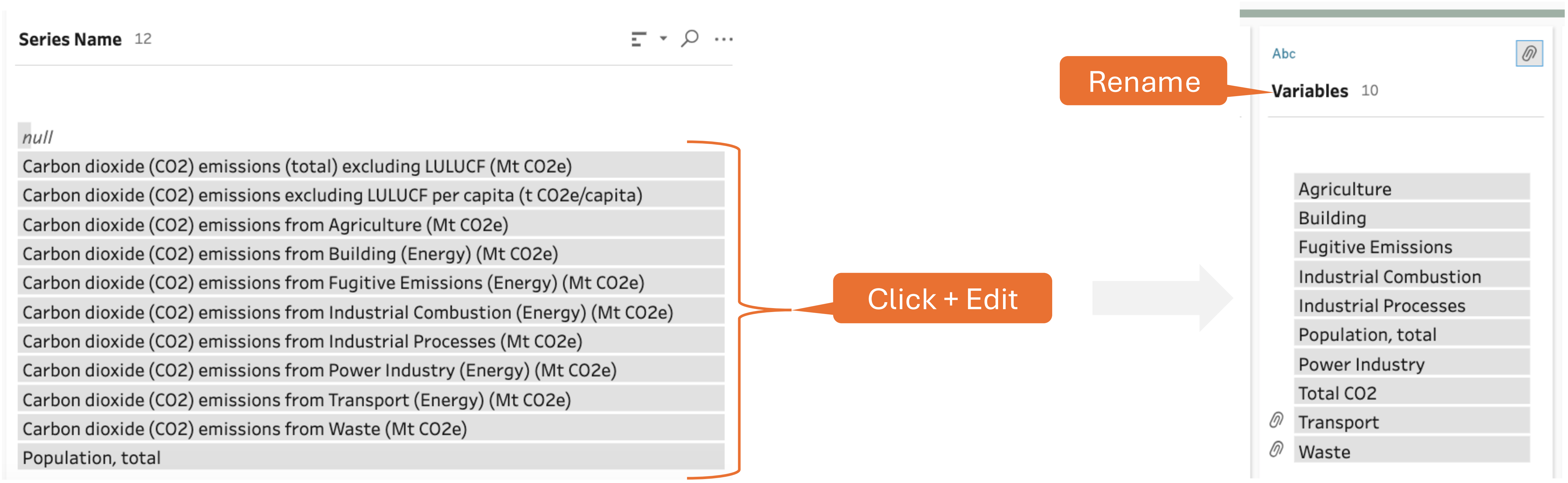
Mantra. Keep the simple things simple!

Strategies.

1. **Click and Edit for small changes**
2. Use the MENU options when convenient.
3. Only resort to code when necessary

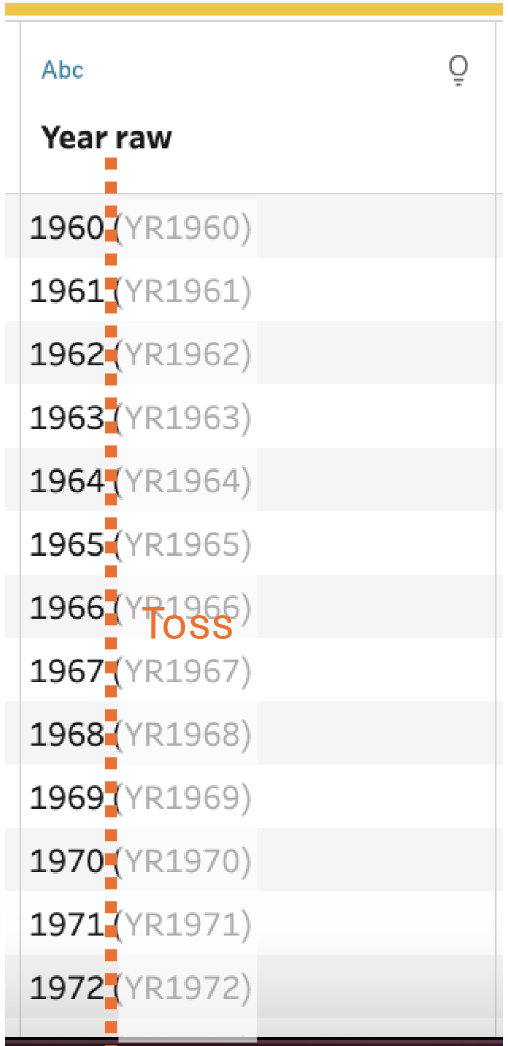
Goal: Clean up the VARIABLE names

Solution: Click and edit the labels



Goal: Clean up the YEAR RAW column

Solution: Split the string



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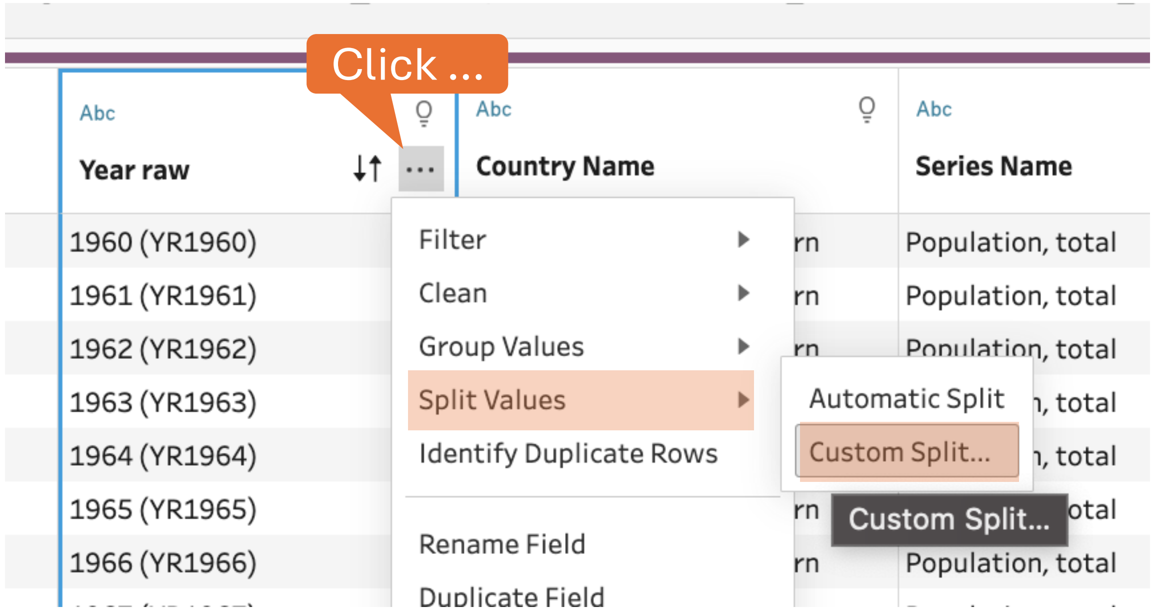
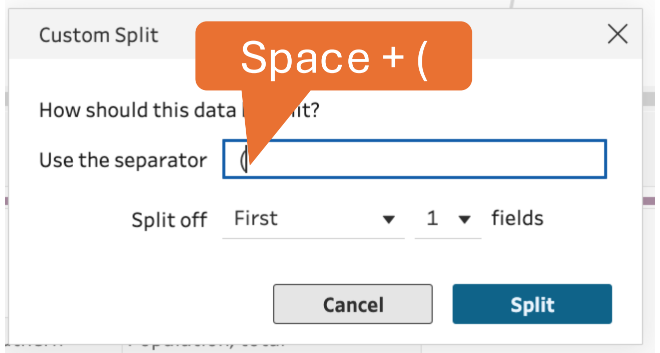
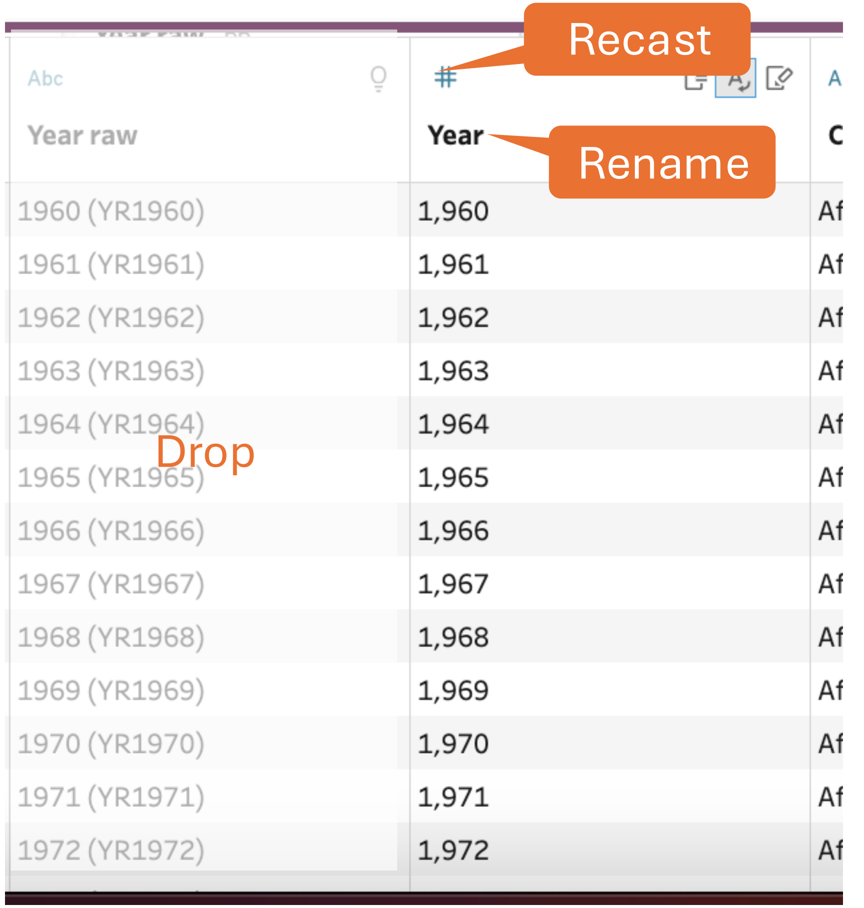
Cleaning up the YEAR column

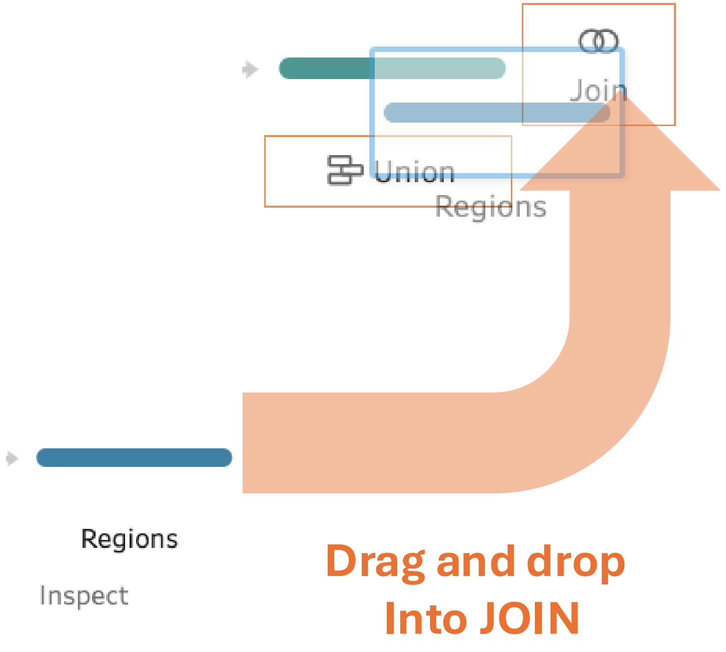
Split → Rename → Cast as whole number → Drop YEAR RAW







JOIN on the region names



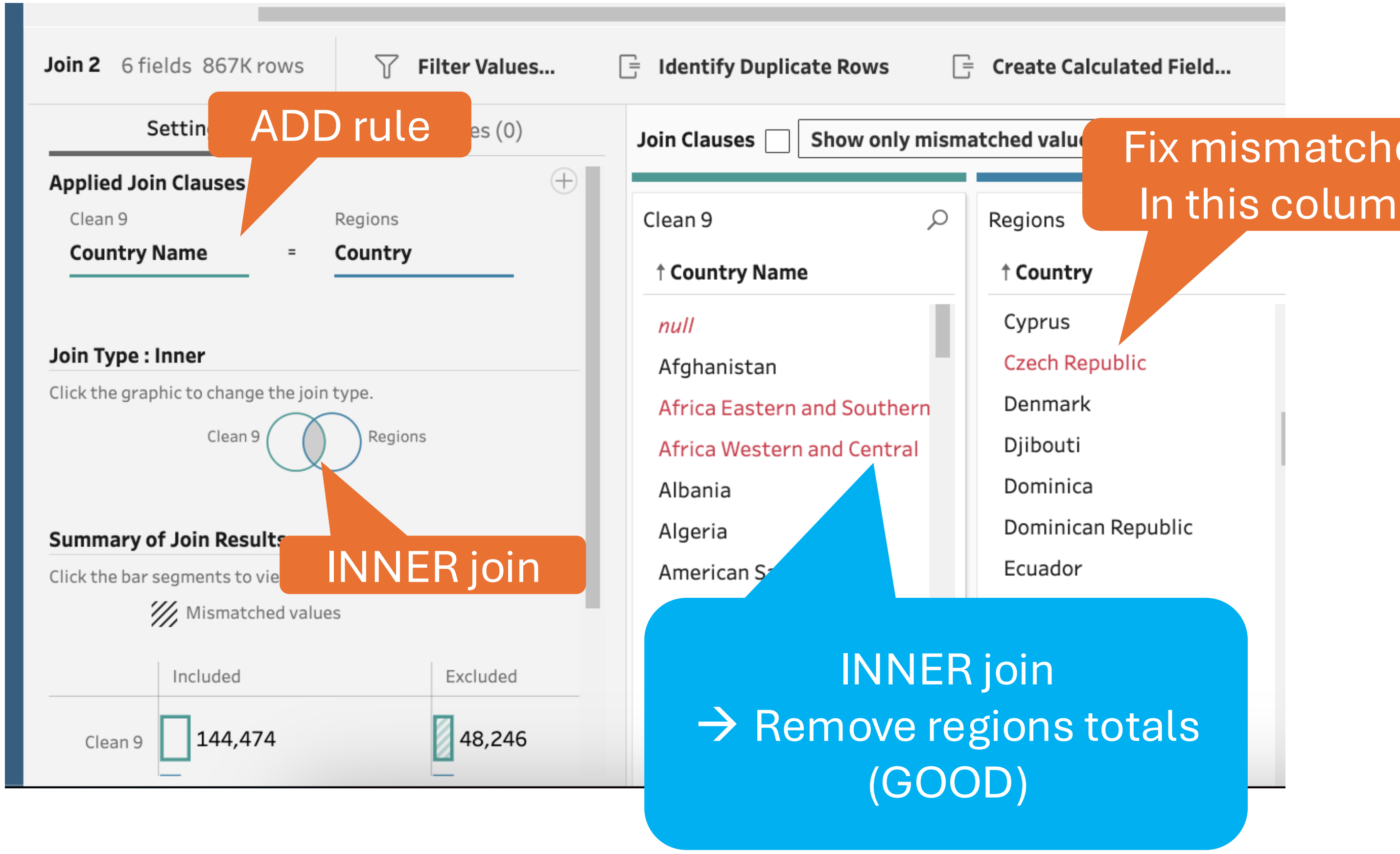
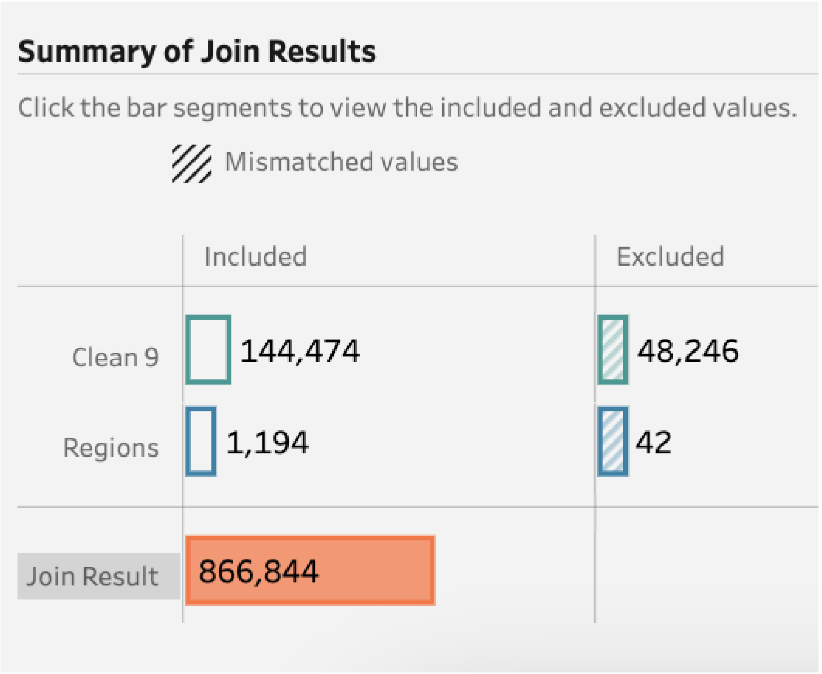
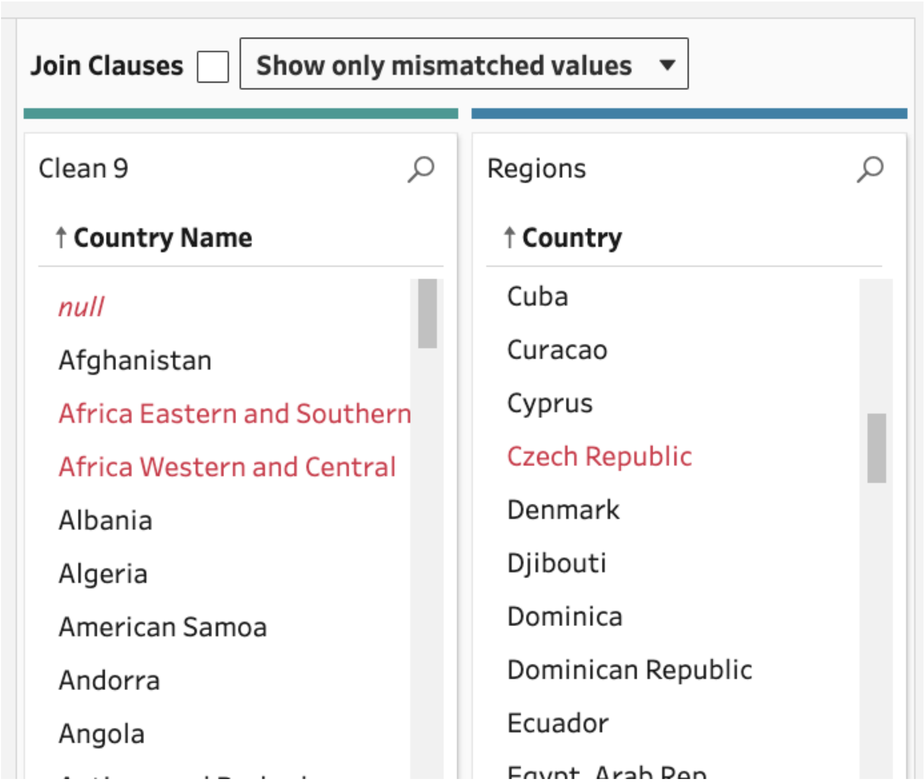


Tableau Prep → Joins done right!

Great summary of results



Find & Fix Mismatches



Goals

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   1. **Goal: Create a longitudinal line graph**
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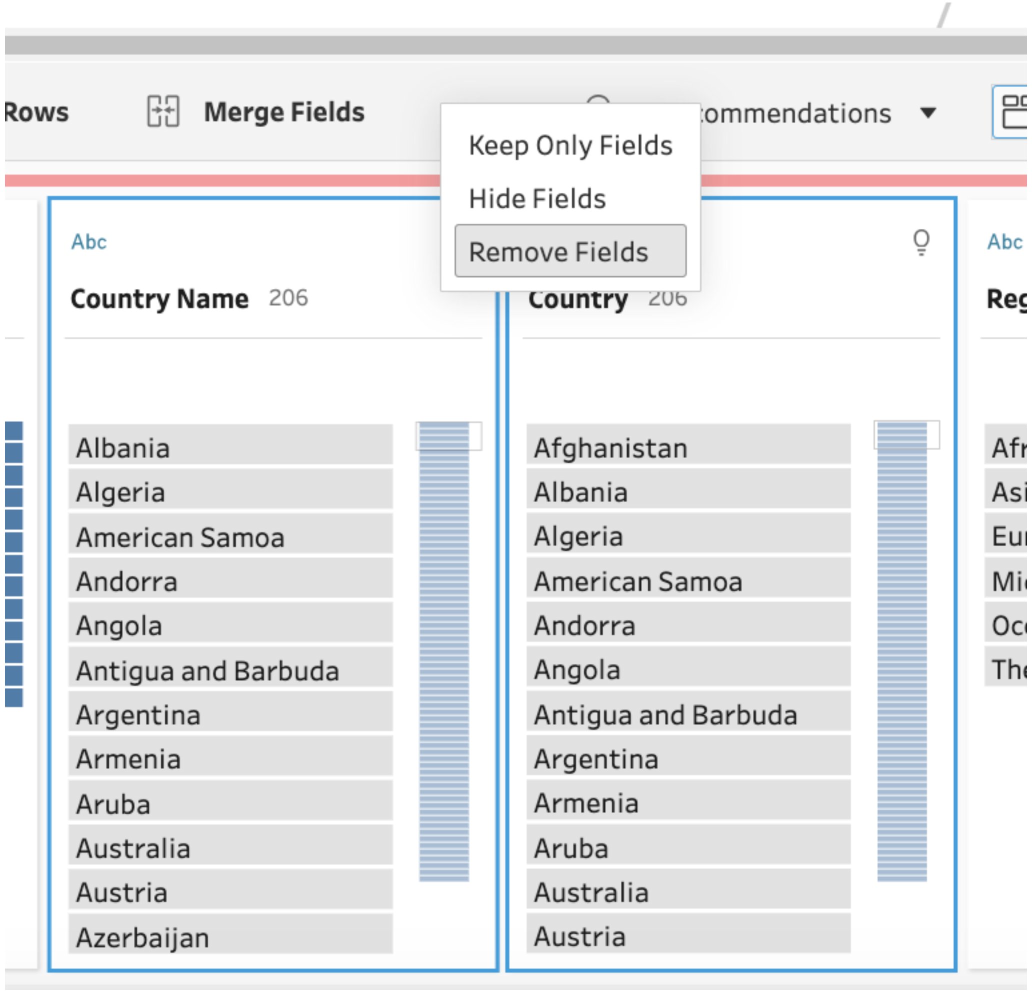
Goal: Compute CO_2 per 1000 people per year + region + source

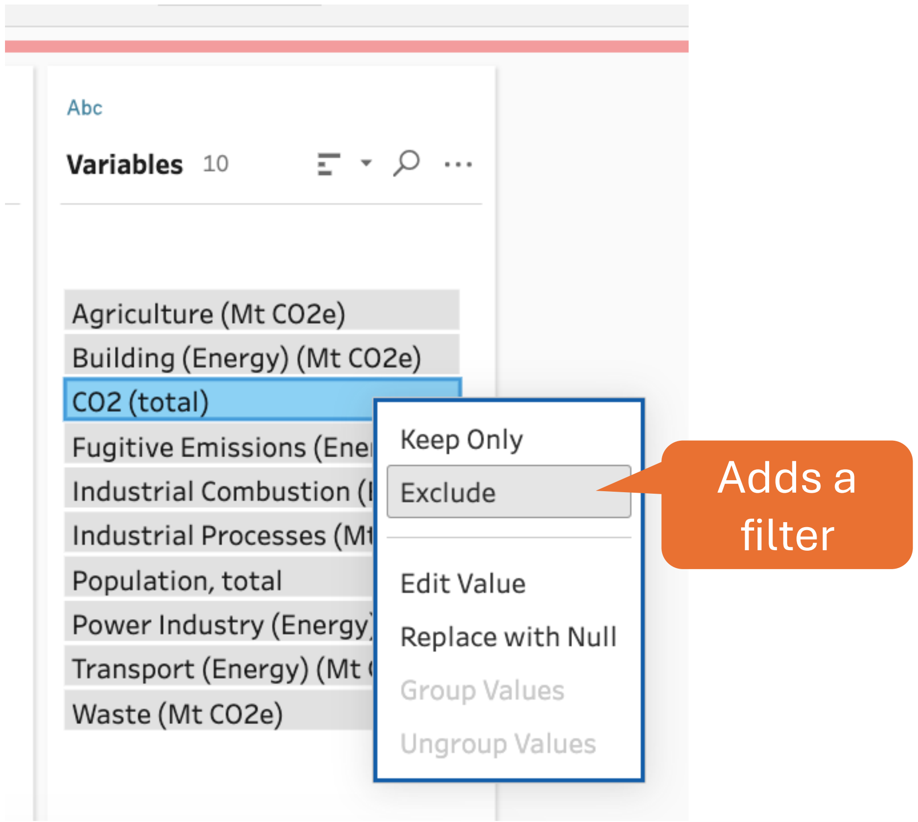
We group by region

→ Drop the country names



Goal: Compute CO_2 per 1000 people per year + region + source

Filter out the total emissions



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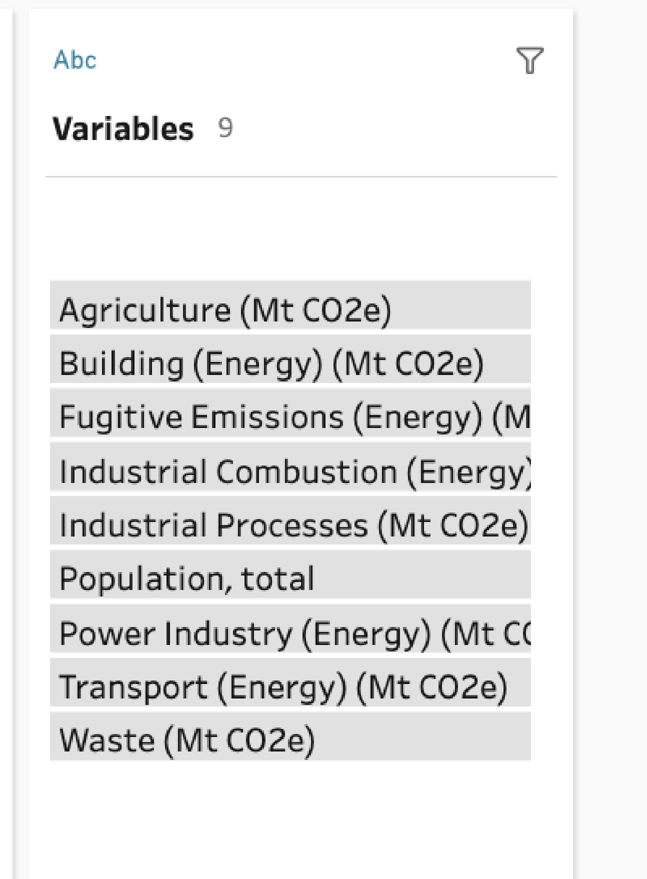
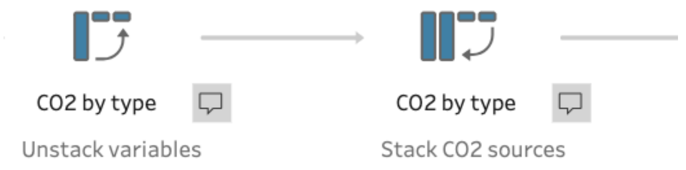
Goal: Compute CO2 per 1000 people per year + region + source

We have a problem

Problem. We need to pull the population totals into a column to compute CO2 per 1000 people





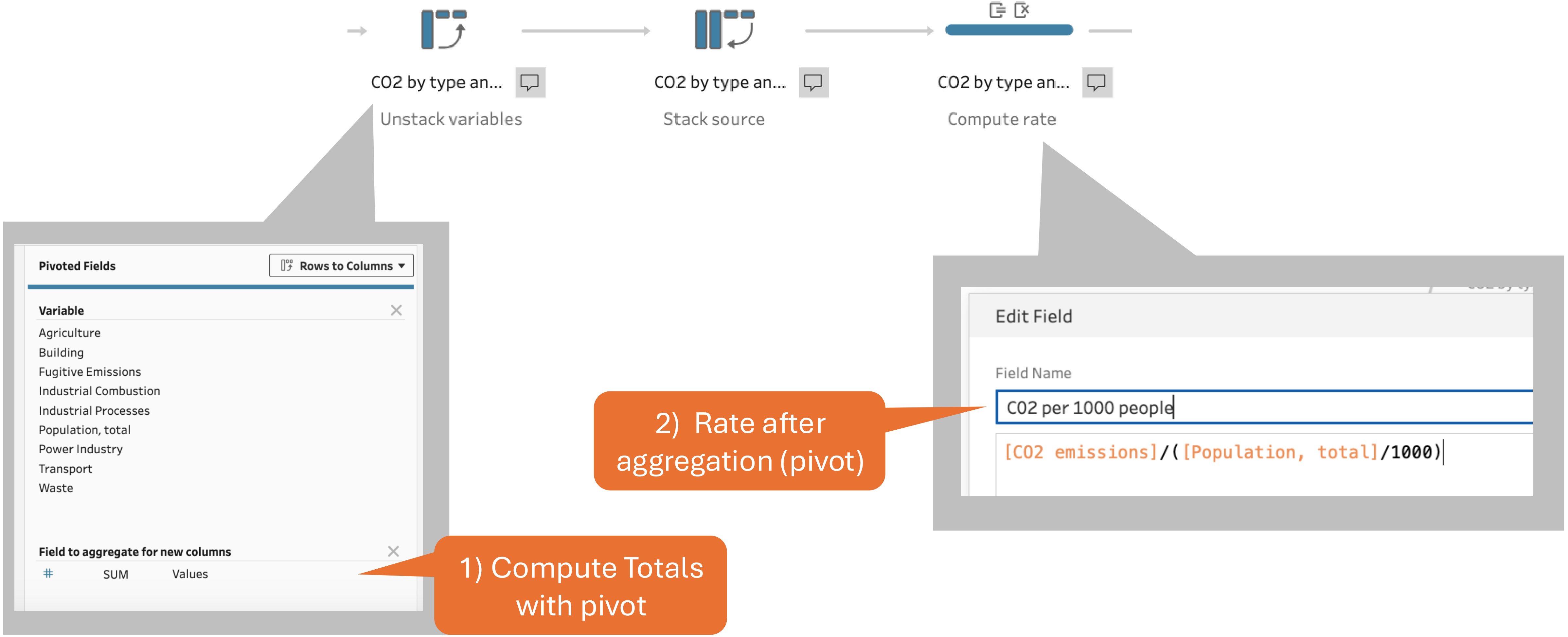
Abc Type	# Population, ...
Agriculture (Mt CO2e)	21,801,040,170
Building (Energy) (Mt CO2e)	21,801,040,170
Waste (Mt CO2e)	21,801,040,170
Industrial Combustion (Energy) (Mt CO2e)	21,801,040,170
Fugitive Emissions (Energy) (Mt CO2e)	21,801,040,170
Industrial Processes (Mt CO2e)	21,801,040,170
Power Industry (Energy) (Mt CO2e)	21,801,040,170
Transport (Energy) (Mt CO2e)	21,801,040,170
Industrial Processes (Mt CO2e)	1,146,638,208
Fugitive Emissions (Energy) (Mt CO2e)	1,146,638,208
Building (Energy) (Mt CO2e)	1,146,638,208

Solution. Unstack Variable → Stack CO2 sources

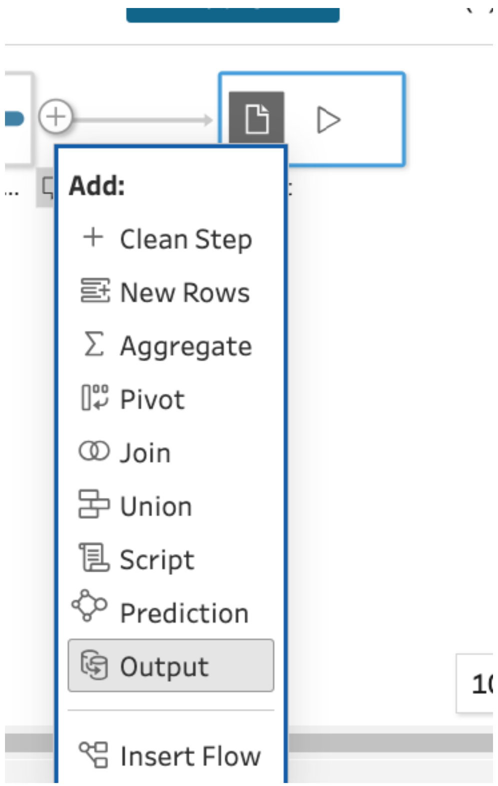
Goal: Compute CO2 per 1000 people per year + region + source

Computing a per-region rate

Compute totals → Divide



Outputting a CSV file



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Drop the country names

Goal 1: Compute the top 10 overall combinations (year + region)
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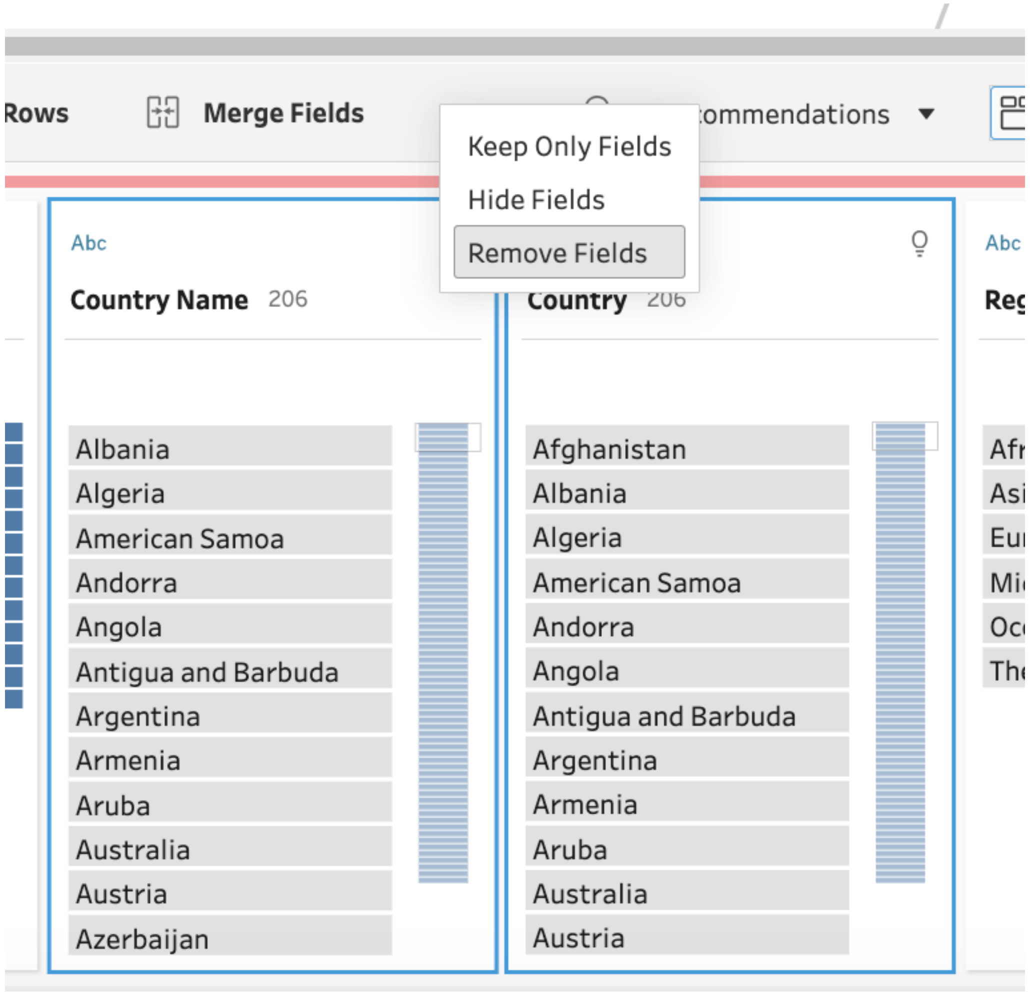

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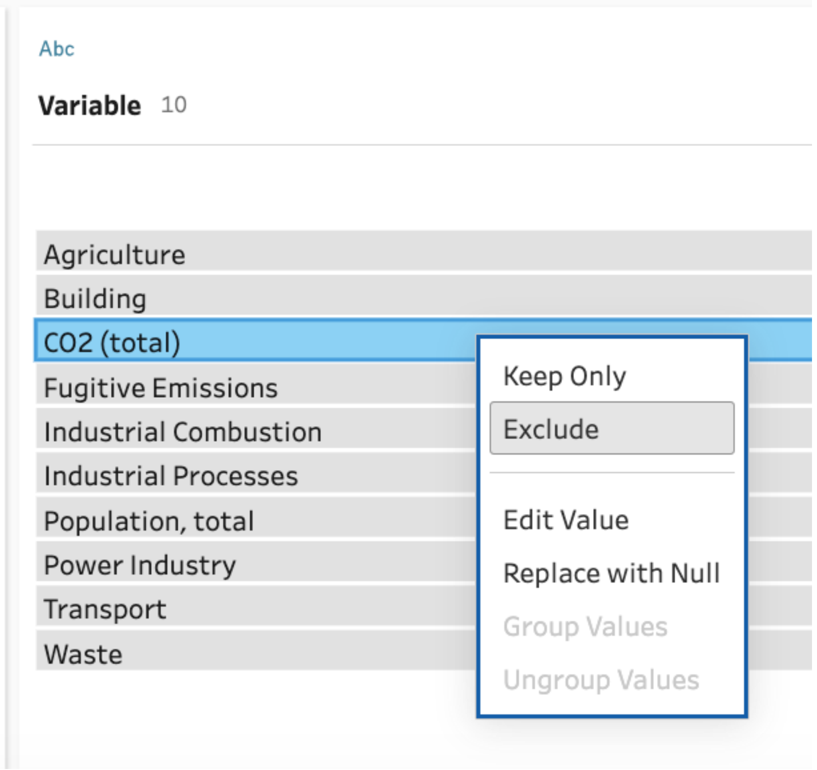
Filter out the total



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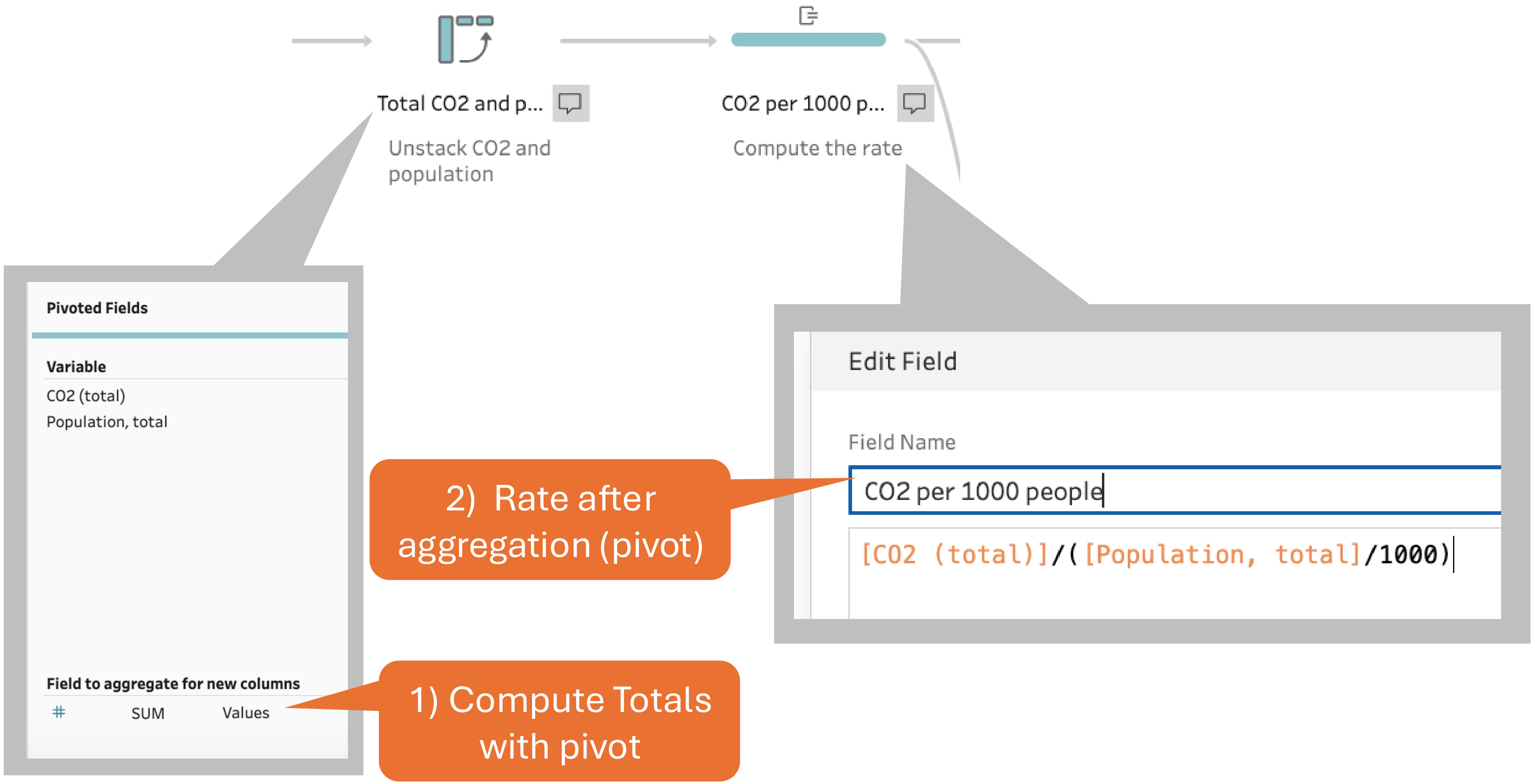

Split → Rank → Filter

Goal 1: Compute the top 10 overall combinations (year + region)
Goal 2: Compute the top 3 regions per year for 2018+